## Cyclic Paths :

### Problem

- Sequential circuit contain feedback (cyclic) paths.
- These paths can cause race conditions and unstable behaviour
- Analysing such circuits may lead to design errors.

### Solution :

- Insert registers in the feedback system.
- Registers store the system state and update only on the clk edge.
- This converts design to combinational + registers, eliminating races.

## Rules of Synchronous Sequential Circuits

- Every circuit element is either a register or combinational logic.
- The circuit contains at least one register.
- All registers share the same clk signal.
- Every feedback (cyclic) path contains at least one register.

X ——— X ——— X ———